

RESEARCH ARTICLE

Exhaustivity and contrastivity in Hungarian focus constructions: A visual-world study

Author¹ and Author¹

¹University, Address, City, Country.

Received: 01 Jan 2026; **Revised:** 02 Jan 2026; **Accepted:** 03 Jan 2026

Keywords: Hungarian, syntax, focus, exhaustivity, contrastivity, visual-world paradigm, sentence–picture verification, self-paced reading, eye-tracking

Abstract

Abstract goes here. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Contents

1	Introduction	2
1.1	Background	3
2	Experiment	6
2.1	Participants	6
2.2	Materials	6
2.3	Design	8
2.4	Procedure	9
2.5	Analysis	10
3	Results	12
3.1	Behavioral results	12
3.2	Reaction Times	13
3.3	Eye-tracking results	21
4	Discussion	22

4.1 Default interpretation of PVF: strong exhaustive bias, but defeasible 23

4.2 Why do exhaustive choices persist under *többek között*? 24

4.3 Processing locus: verification conflict rather than incremental sentence decoding 24

4.4 Implications for the exhaustivity debate 25

4.5 Limitations and future directions 25

4.6 Interim conclusion 26

5 Conclusion 26

6 NOTES AND MISC 26

6.1 TODO 26

1. Introduction

How languages express contrast and exhaustivity has major implications for how we understand the syntax–semantics interface. This study revisits a theoretical debate on the interpretation of Hungarian focus constructions using eyetracking technology and the visual-world paradigm. Specifically, we pursue two theoretically important inquiries: what the dominant and available readings of Hungarian pre-verbal focus constructions are, as well as from a cognitive perspective, whether different focus types are processed differently during language comprehension.

- (1) *Anna meg-evett egy banánt.*
Anna[NOM] VPX-eat-PST-3SG a banana-ACC
'Anna ate a banana.'
- (2) *Anna egy BANÁNT evett meg.*
Anna[NOM] a banana-ACC eat-PST-3SG VPX
'Anna ate a BANANA.'

Consider the canonical, neutral sentence in example (1)¹, where the subject *Anna* is in the sentence-initial position, followed by the verb, *evett* 'ate' (together with the verbal prefix [VPX] (or verbal modifier [VM]) *meg-*)² and the object *banán* 'banana' (with the *-t* accusative ending); vs the pre-verbal focus construction in example (2), where the object is fronted to the pre-verbal position, leaving its (normally attached) verbal prefix behind. The sentence in example (2) is a typical example for a Hungarian focus construction, and it could also be rendered in English with an *it*-cleft,³ i.e., 'It is a BANANA⁴ that Anna ate.'

Constituents in canonical sentences can be focalized for various communicative purposes, such as emphasis, question answering, or contradiction. In (2) we are focusing the object, and as such we can observe two changes that mark focus: movement and prosody. Regarding the first, Hungarian has a special, grammaticalized syntactic position for the

¹Gloss label abbreviations in this and all further examples – except VPX=verbal prefix – are from the Leipzig Glossing Rules (Comrie et al. 2015): ACC=accusative, DEF=definite, NOM=nominative, PL=plural, PST=past tense, REL=relative, SG=singular.

²Verbal prefixes in Hungarian can modify the meaning of the verb, or act as aspect markers. *Meg-* in this case has a perfective aspect meaning, indicating not only that the action has been concluded but that the event is complete, with the apple wholly eaten.

³While the comparison of the Hungarian focus constructions with English *it*-clefts has received considerable attention in the literature (cf. É. Kiss 1999; Wedgwood et al. 2006) it falls outside the scope of the present discussion.

⁴Linguistically focused elements in the examples will be set in capitals for highlight throughout the paper.

focused constituent directly in front of the verb – called the pre-verbal focus (PVF) position (Balogh 2007; Wedgwood 2006) or structural focus position (É.Kiss 1998) or simply just focus position – often associated with exhaustive/identificational semantics (Balogh 2013). Besides movement, the syntactically expressed focus is also accompanied by prosodic stress, and it has been suggested that in these focus marking strategies movement is primary, and prosody is secondary (Balogh 2013). Regarding intonation in PVF constructions, the focused constituent typically bears eradicating stress (cf. Gécseg 2020; Kornai & Kálmán 1988), meaning that the verb is destressed and the acoustic prominence of all other elements are leveled. The PVF construction is a common, everyday focus-marking strategy in Hungarian – a topic-prominent, discourse-configurational language where the configuration of constituents expresses information structural roles as opposed to grammatical ones (cf. É. Kiss 1995) – and various aspects of it has garnered significant attention in both theoretical and psycholinguistic research.

1.1. Background

One of the key issues around Hungarian pre-verbal focus is whether it is inherently exhaustive, or if the exhaustivity is a pragmatic inference. In other words, does the PVF construction itself entail that the focused element is the only one for which the predicate holds true, or can it be interpreted in a contrastive way, where the focused element stands in opposition to other alternatives without necessarily excluding them? For example, in (2) above, does this sentence implies that Anna ate only a banana *and nothing else* – an exhaustive reading, or could it mean that Anna ate a banana *and not something else* (or even, that she ate a banana but maybe *she also ate something else*) – a contrastive reading? We are interested in the ‘default’ interpretation of focus constructions such as these, and whether Hungarian speakers tend to understand them exhaustively or contrastively without additional context.

We should therefore try to define what we mean by *exhaustive focus* and *contrastive focus*. First, both types require the pre-verbal structure, but they differ in their semantic/pragmatic implications. In É.Kiss (1998)’s influential paper, both would fall under the term of *identificational focus*, which she defines as a focus that identifies the referent of the focused element from a set of alternatives. It takes scope, binds a variable, and interacts with quantification. É. Kiss treats contrastive focus as a subtype of identificational focus, where she describes it as [+exhaustive], and often [+contrastive]. However, we will argue that identificational focus does not always entail exhaustivity. According to Rooth (1992)’s theory on focus interpretation, exhaustivity is not inherent to focus but arises when focus interacts with specific operators like ‘only’. For him, focus is a mechanism that introduces a set of alternatives (*alternative semantics*), and he considers contrast to be the pragmatic use of focus, where one alternative is highlighted against others in the set. In Vallduví & Vilkkuna (1998), *kontrast* is an operator-like category that introduces a set of alternatives for the focused constituent. If an expression *a* is contrastive, a membership set $M = \{\dots, a, \dots\}$ is generated, serving as a quantificational domain. This allows alternative propositions $P(x)$ to be derived by substituting other members of *M* for *a*. Thus, *kontrast* provides the semantic mechanism by which focus evokes alternatives, enabling identificational or exhaustive readings. In Krifka (2008:259)’s work on Information Structure (IS), both exhaustive and contrastive focus is mentioned as subtypes of focus, and while exhaustive is interpreted as the only alternative that makes the proposition true narrowing the set of

alternatives down to one (and thus enforcing exclusivity), contrastive focus is defined as a focus used for truly contrastive utterances and presupposes a competing alternative in the common ground, highlighting the contrast. Krifka's distinction aligns with his broader distinction between semantic (truth-conditional impact) and pragmatic (discourse management) uses of focus. Zimmermann (2008) also argued that contrastivity is not a semantic property of focus at all but rather a pragmatic one, and that it is best understood in terms of hearer expectation and discourse predictability, as a discourse-semantic device that signals unexpectedness relative to the hearer's assumptions, rather than a purely semantic operator over alternatives.

In short, exhaustive focus implies that the focused element is the sole entity for which the predicate holds true; i.e. that there are no other things Anna ate. On the other hand, contrastive focus would imply that the focused element stands in opposition to other possible alternatives, drawing attention to its uniqueness, unlikelihood, or contradicting nature given the context, e.g., 'Anna ate a banana [and not an apple].', but it does not necessarily exclude alternatives, i.e., there could be other things that Anna ate, e.g., 'Anna ate a banana [and/but also an apple].' Now consider the following:

- (3) Anna *csak* egy BANÁNT evett meg.
 Anna[NOM] only a banana-ACC eat-PST-3SG VPX
 'Anna ate (up) only a BANANA.'
- (4) Anna *többek között* egy BANÁNT evett meg.
 Anna[NOM] others amongst a banana-ACC eat-PST-3SG VPX
 'Anna ate (up), amongst other things, a BANANA.'

We can, in fact, force an exhaustive reading by adding the exhaustive focus marker *csak* 'only' to the sentence, as in example (3)⁵. One intuitive reading of a simple PVF sentence like that in example (2) would be that 'Anna ate a banana [and nothing else].' In this sense, the sentence seems to be equal to that in (3), where the exhaustive interpretation is explicitly expressed with the operator *csak* 'only'. If we accept É.Kiss (1998)'s view, this seems to be redundant as the structural focus is already exhaustive. The question of differentiating the exhaustivity entailed by the pre-verbal focus structure vs the exhaustivity operator has been studied in detail by Balogh (2007).

Moreover, we can also cancel exhaustivity within the same clause with the help of certain phrases, such as *többek között* 'amongst others', as in example (4)⁶, by inserting a phrase with an inherently non-exhaustive, inclusive meaning (see also Balogh 2013; Wedgwood 2006:14). This sentence is not only perfectly grammatical, but inclusive of alternatives and expresses a contrastive sense due the use of focus (there must be some specific discourse function for choosing to focus a constituent).

In Hungarian linguistic research, the role of semantic exhaustivity in focus constructions has been debated for more than forty years. The central question is whether exhaustivity arises from the syntax–semantics interface or from pragmatic and contextual factors. Earlier accounts within the generative framework assumed that the Hungarian pre-verbal focus structure necessarily entails semantic exhaustiveness, claiming that exhaustivity is an

⁵The scope of *csak* 'only' only extends to the subsequent element, it can only refer to the object and not the following VP.

⁶Similarly, only the object is in the scope of *többek között* 'amongst others', it is not possible to interpret this sentence as if the verb is an alternative to other actions one can do to a banana.

inherent property of the focus structure itself (É.Kiss 1998, 2007; Horvath 2008; Kenesei 2006; Szabolcsi 1981), which was more or less the received view up until the 2000s. More recently, this claim has since been challenged and shown to be untrue by both theoretical and psycholinguistic work – supported by experimental data – and it has been proposed that the exhaustive interpretation emerges from discourse-pragmatics, or conversational implicature (Balogh 2013; Káldi and Babarczy 2016; Onea 2009; Wedgwood et al. 2006). In introducing their argument against the claims of Szabolcsi (1981), É.Kiss (1998, 2007), and Horvath (2008) that ‘an immediately pre-verbal focus in Hungarian is semantically exhaustive’ or ‘is interpreted exhaustively, i.e. as if it were in the scope of an exclusive like “only”’, Onea & Beaver (2009:1) suggested in their first example⁷ that the implicit reading of a sentence of this type should be exhaustive, i.e. ‘Peter kissed Mari and noone else.’. Their example is analogous to our example (2), i.e. ‘Anna ate a banana and nothing else.’, and their analysis suggests that while such sentences often invite an exhaustive reading, this interpretation is not structurally encoded but contextually inferred.

Thus, looking back at example (2), the underlying implication is not necessarily that Anna ate *nothing* else, but it could be the case that she did not eat *something* else, as shown in (5a). Therefore, in our view, the interpretation might as well be contrastive. In other words, those who claim that only one of the readings is available to PVF construction would expect only one of the two readings to survive, contrary to the facts.

- (5) a. Anna egy BANÁNT evett meg, és semmi mást.
 Anna[NOM] a banana-ACC eat-PST-3SG VPX and nothing else-ACC
 ‘Anna ate a BANANA and nothing else.’
 b. Anna egy BANÁNT evett meg, nem egy almát.
 Anna[NOM] a banana-ACC eat-PST-3SG VPX and not an
 ‘Anna ate a BANANA and not an apple.’

To test this, and to address the theoretical debate, we designed an experiment to investigate how Hungarian speakers interpret pre-verbal focus (PVF), and in particular whether they distinguish between exhaustive and contrastive focus in line with recent experimental studies (Gerőcs et al. 2014; Káldi et al. 2021; Onea & Beaver 2009). Specifically, we aim at

⁷ Péter MARIT csókolta meg. Peter[NOM] Mari-[ACC] kiss-PST-3SG.DEF VPX ‘Peter kissed MARY.’

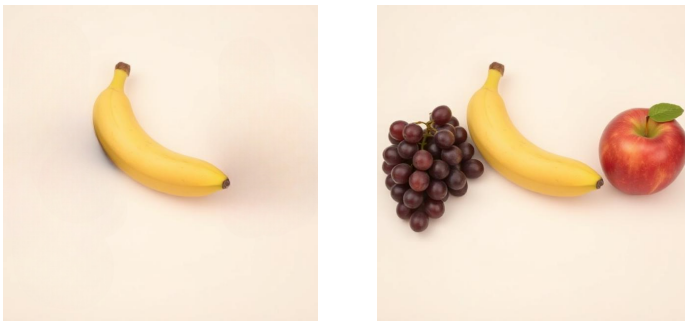


Figure 1. Example stimuli for a sentence–picture verification task

answering the following questions: What is the default interpretation of a pre-verbal focus construction? Is it really inherently exhaustive, or could it be contrastive?

2. Experiment

We designed a visual-world experiment with eyetracking in which participants first read a sentence and then chose between two plausible pictures: (A) a single-object image corresponding to an exhaustive interpretation, or (B) a multi-object image corresponding to a contrastive interpretation (see Figure 1 and examples (1)–(4)). Our main question is how unmodified PVF sentences (e.g., (2)) pattern relative to the *only*–exhaustive condition (ex. (3)) and the *amongst others*–contrastive condition (ex. (4)). We expect *only* sentences to yield predominantly exhaustive choices and *amongst others* sentences to favor contrastive choices; the crucial issue is whether simple PVF sentences align with one of these conditions or show a more balanced distribution. In addition to the choices, we record reaction times and collect eye movements during picture verification in order to assess whether the conditions have an impact on cognitive processing, and whether different interpretations are associated with different reaction times and/or patterns of attention.

We hypothesize that *only*–exhaustive sentences will be associated with single-object (A) images, with faster responses and more focused attention, while when presented with *amongst others*–contrastive sentences, participants will prefer the multi-object (B) images and may exhibit slower responses and more distributed attention due to the presence of alternatives and the ambiguity of the underlying meaning. This would suggest that the PVF construction in itself is not exhaustive.

2.1. Participants

We recruited 55 native speakers of Hungarian via the Prolific platform (www.prolific.com), where participants were pre-screened for Hungarian as their first language, and received a monetary compensation for their time (around £7.50/h). After rejecting 3 participants (due to overly long submission time, failing attention checks, or missing ET data), the final sample consisted of 52 individuals (32 female; mean age = 35). On average it took people just under 23 minutes to complete the experiment, although with a high individual variation (unusually long submission times were rejected).

2.2. Materials

Our stimuli consisted of 54 experimental sentences (18 items × 3 conditions) with 54 additional filler sentences in a randomized order (108 in total), but making sure that the same items never followed each other. Regarding lexical variation, we used verbs both with and without verbal prefixes (50-50%), and we included instances of both definite and indefinite conjugation (44.4-55.6%); also, both countable and uncountable nouns were used for object focus. We added some simple contextual fillers to the beginning of each sentence, such as *ami X-t illeti* ‘regarding X’ and time or place adverbials to balance sentence length and make the targeted focus constructions appear not in the beginning of trials. When presenting the sentences on the screen, we divided them into regions, and articles were kept together with nouns, so all three conditions had the same number of regions (7), and the

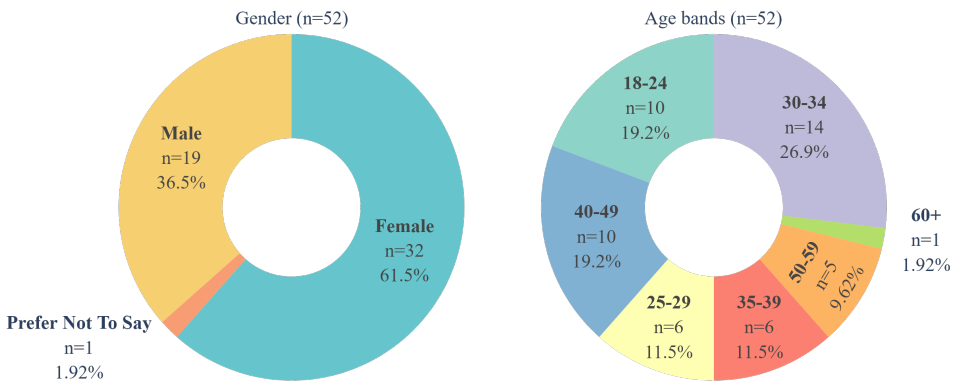


Figure 2. Demographic information of the participants in the experiment; gender (left) and age distributions (right)

focused element was always in region 6 (r6). The stimuli were created by modifying a set of 18 base sentences – which were all simple transitive sentences – by modifying the base sentence with the addition of the focus marker *csak* ‘only’ for the exhaustive condition, and the phrase *többek között* ‘amongst others’ for the contrastive condition.

The 18 items were accompanied by 18 image pairs, these were created using DeepAI’s image generator (<https://deepai.org/>) with prompts requesting simple 3D objects & scenarios in front of a light pastel background, and then modified manually to get matching pairs. The full stimuli list is available online,⁸ and in the Appendix (see the sentence list in Table 6.1 and the image pairs in Fig. 12).

To avoid overly long experimental sessions, we have split the experiment into three equal parts and each participant could complete one, two, or all three parts as separate events; meaning that we had three participant groups that could overlap, and each experiment consisted of 40 trials (36 experimental + 4 attention checks). This resulted in 72 valid submissions (25 for group one, 23 for group two, and 24 for group three), thus every stimuli was interacted with at least 23 times. The three parts were split in a way that for each group each item only appeared in one condition, and each session contained four attention check questions intermixed and evenly distributed across the trials.

The three conditions were: a) exhaustive focus with *csak* ‘only’ (e.g., (6a)), b) unmodified, simple pre-verbal focus (PVF) (e.g., (6b)), and c) contrastive focus with *többek között* ‘amongst others’ (e.g., (6c)). Each item was paired with two images: A) one depicting a single-object interpretation – exhaustive reading – and B) one depicting a multiple-object interpretation – contrastive reading. See Figure 3 for an example of the image pairs accompanying the sentences in the experiment.

(6) a. **exhaustive focus**

Ami az innivalót illeti, Eszter csak kávét ivott.
 which[REL] the drink-ACC regard-3SG Eszter only coffee-ACC drink-PST-3SG
 ‘Regarding drinks, Eszter only drank coffee.’

⁸OSF link: https://osf.io/j9abg/overview?view_only=484a4f21a5cb4e51b4b826f970e096a

b. **unmodified pre-verbal focus**

Ami az innivalót illeti, Eszter ma kávéét ivott.
 which[REL] the drink-ACC regard-3SG Eszter today coffee-ACC drink-PST-3SG
 ‘Regarding drinks, Eszter drank coffee today.’

c. **contrastive focus**

Ma reggel Eszter többek között kávéét ivott.
 today morning Eszter others amongst coffee-ACC drink-PST-3SG
 ‘This morning, Eszter drank, amongst other things, coffee.’

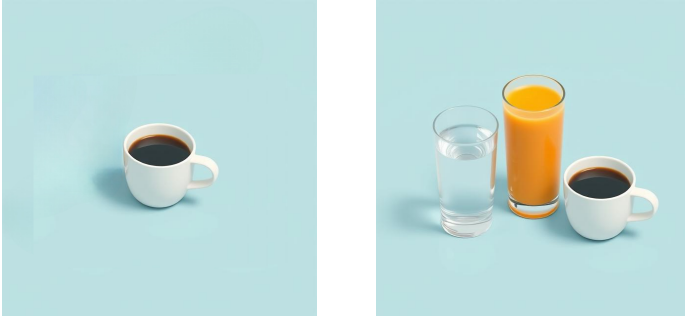


Figure 3. Image pair for the sentence in examples (6a), (6b), and (6c). The left image (A) corresponds to the exhaustive reading, while the right image (B) corresponds to the contrastive reading.

2.3. Design

We built a sentence–picture verification (SPV) task, in which participants read a sentence and were asked to indicate the best fitting meaning-representation between two visual stimuli. First, participants saw a preview of the image pair to get familiar with the objects/context, on the left and right half of the screen (20% of the screen’s width and 40% of the screen’s height) on a canvas 50% larger than the images to mitigate the inaccuracies of the eye tracker. After pressing the spacebar they were presented with the sentence word by word (region by region) in the middle of the screen in a self-paced reading (SPR) manner, in which participants had to use the spacebar to advance through the sentences. We recorded reading times throughout the reading. After reading through the sentence, the image pair re-appeared and participants were asked to select the image that best matched the meaning of the sentence they just read. The choice had to be made with a mouse-click anywhere on the corresponding canvas (left or right), and reaction times (RT) and eye movements were recorded during this picture verification phase. The image types were counterbalanced throughout the trials. The segments were separated by 1000 ms pauses, and after the choice, the next trial started after a brief (1000 ms) pause. The experiment was designed to be completed in around 25-30 minutes, and participants had to complete it in one setting. See Figure 4 for a visual representation of the design of the experiment.

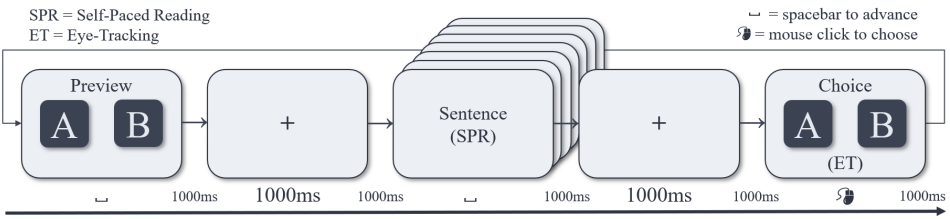


Figure 4. Design

2.4. Procedure

The experiment was created, hosted on, and presented (in Hungarian) to participants on PCibex Farm (Schwarz & Zehr 2021).⁹ Participants completed the experiment on their own devices (PC only, with a webcam and physical keyboard & mouse) and on their own time, but with Prolific helping us control for too fast or too slow submissions resulting in rejection or timeout.

First, potential participants were instructed to be in a well-lit environment, sit at an appropriate distance from their screen and webcam, and minimize head movements, emphasizing the need for a bright environment, and that they can only start the experiment from a Chrome browser. Before starting the experiment, participants then had to calibrate the eye tracker, which involves looking at specific points on the screen (e.g., corners, center) while the system records their eye positions to create a mapping between gaze coordinates and screen coordinates. Eye-tracking data were collected using the built-in WebGazer.js library of PCibex, which uses the participant's webcam to track eye movements, and we used the default calibration procedure provided (a 9-point calibration grid). Participants had to reach and maintain at least 60% accuracy in the calibration phase to proceed with the experiment, and had three attempts to achieve this accuracy, ensuring that the eye-tracking data collected during the picture verification phase would be reliable. The tracker also checked their gaze again and again before every trial, and if the accuracy dropped below 60%, participants were prompted to recalibrate before proceeding to the next trial. People who could not pass the calibration for various reasons (e.g., computer setup, bad lighting/environment, glare on glasses, or any other technical issue) were not able to start the experiment. After calibration, participants were instructed on the exact task and how to complete the trials, followed by a practice session consisting of three trials, and finally the live experiment of 40 trials (including 4 attention checks).

We have to mention that a relatively large amount of people (around a third of all who tried) could not pass calibration, and thus were not able to start the experiment. This is a persistent issue with webcam-based online eye tracking studies, as it can be affected by various factors such as lighting conditions, webcam quality, and even individual differences in facial features. Depending on these factors, participants' experiences were vastly different, some people breezed through trials and had a smooth experience finishing under 20 minutes, others struggled and had to recalibrate from time to time, spending over 30-40 minutes in total. We urged participants to reach out if they had issues, but sadly we

⁹Code and resources are also available online for inspection: https://osf.io/j9abg/overview?view_only=484a4f21a5cb4e51b4b826f970e096a

couldn't assist everyone remotely. A small number of participants started but couldn't finish the experiment due to technical reasons, they were also compensated for their time, but their data was not included in the analysis.

2.5. Analysis

2.5.1. Behavioral data?

2.5.2. Eye-tracking data processing

Eye-tracking data recorded on PCIBex comes in a relatively straightforward, clean format not of gaze coordinates but in classified samples, where each sample is already defined as a gaze at a canvas or not, in our case: left canvas, right canvas, or neither. Samples are recorded between every 30-70 milliseconds, depending on the users' computer & browser setup, internet speed, and presumably the general load that the PCIBex server was under at the time.

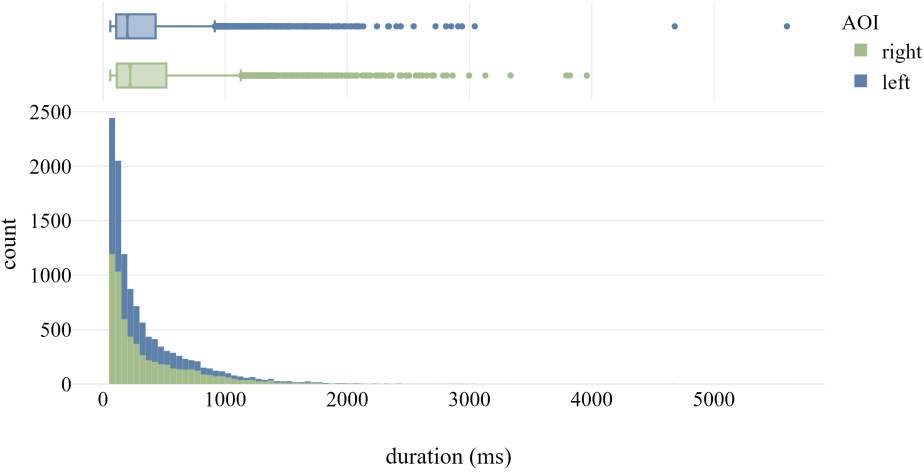


Figure 5. Distribution of episode durations after cleaning and before winsorization

The following steps were taken to clean and classify the eye-tracking data in order to obtain a certain set of metrics that could be used for analysis: First, raw eye tracking data files were loaded per participant, and **samples** were classified into AOIs at each timestamp as left (left canvas=1, right canvas=0), right (right canvas=1, left canvas=0), or neutral otherwise. Then, same-side consecutive samples were collapsed into contiguous gaze **episodes** within participant and trial. Episode duration was computed from timestamp-derived sample durations where episodes shorter than a threshold of 60 ms were removed, and neutral-gap bridging was enabled (with a max neutral bridge of 50 ms). In other words, very short gaze episodes and rapid saccades were removed cleaning noise & jitter, as they are unlikely to reflect meaningful cognitive efforts and more likely to be artifacts of the

eye-tracking system. Then, neutral episodes were dropped and excluded from further analysis. Episodes thus intend to represent a moment of sustained gaze within an AOI, ignoring brief out-of-canvas samples that result from technical inaccuracies and are not deliberate (or uncounscious) eye movements. For quality control, episode-duration distributions were inspected via summary statistics, and episodes longer than the mean +2 SD ($\mu \pm 2\sigma$) were winsorized by replacing their duration with the global mean (i.e., with rows retained). This procedure preserved the full number of episodes while limiting the influence of extreme outliers; 4.89% of all episodes were substituted. See Figure 5 for the distribution of episode durations. This cleaning process was devised after established eye-tracking data classification protocols, such as the Identification by Dispersion-Threshold (I-DT) algorithm, which itself is ideal for low- to mid-frequency eye trackers and static stimuli.

Finally, trial-level eye-tracking (ET) features were computed, including left/right/total dwell time, dwell proportions, dominant side, fixation counts, revisits, transitions, and first-fixation side/duration. To briefly define the most important metrics that were used in the end: **dwell** is a time value, the sum of the duration of all gaze episodes within a trial (either left, right, or total), representing the total amount of time a participant has spent looking at the AOI(s) during the choice; **fixation** on the other hand is a numeric value, representing how many times a participant looked at (or switched between) the AOIs during a choice. Missing trial-level metrics were mean-imputed in 5.02% of the data points, which were mostly due to missing episode-level data (e.g., no episodes detected for a trial, or no episodes remained after cleaning).

We acknowledge that several processing decisions may limit inference. Gaze data was essentially collapsed to binary AOIs (left/right), while off-target samples were treated as neutral and excluded. Episodes shorter than 50 ms were removed, potentially excluding very brief but real events. Long-duration episodes were winsorized by replacing values above a global mean+2SD threshold with the global mean, preserving sample size but reducing variance and potentially biasing duration-based measures. Also, missing merged ET trial metrics were mean-imputed for selected variables, which further shrinks the effective sample size. We therefore interpret absolute magnitudes cautiously and focus on relative contrasts between conditions.

The eye-tracking data cleaning and classification pipeline was implemented in Python, and can be inspected online.

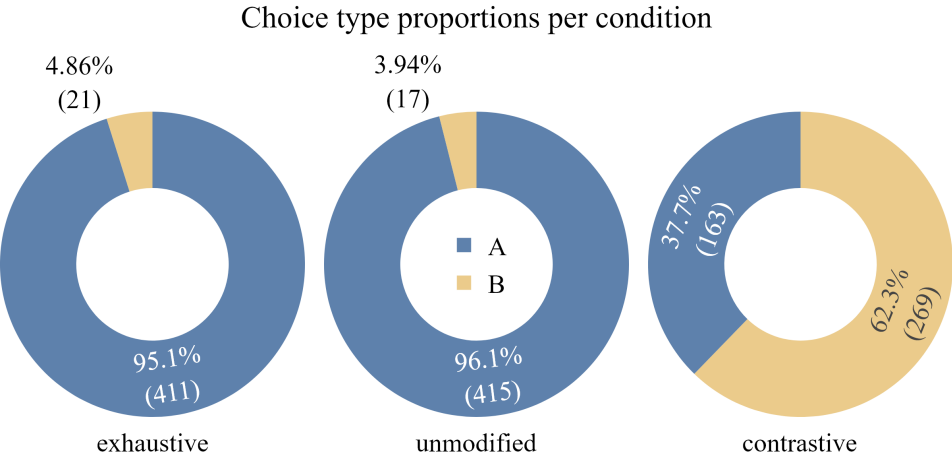
2.5.3. Statistical analysis

We analyzed two eye-tracking dependent measures: *total dwell time* (`total_dwell`) and *total fixation count* (`total_fixation`). For both measures, analyses were conducted in R using linear mixed-effects models (LMM) (packages `lme4`, `lmerTest`, `car`, and `emmeans`). The fixed-effect predictors were focus Condition (three levels: exhaustive, unmodified, contrastive) and image Choice (two levels: single-object, multi-object). To stabilize variance and accommodate zero values, we applied a log transform using $\log(1 + x)$ to the raw dependent measure prior to modeling. For each dependent measure, we fit the following model with Participants and Items as random intercepts: $y' \sim \text{Condition} * \text{Choice} + (1 | \text{Participant}) + (1 | \text{Item})$ where $y' = \log(1 + y)$ is the transformed dependent variable. The model were fit using maximum likelihood estimation (i.e., `REML = FALSE`) to enable likelihood-ratio model comparisons. To follow up on significant effects in the model, we computed estimated marginal means and conducted Tukey-adjusted pairwise

01 comparisons using emmeans. Specifically, we tested (i) pairwise differences between con-
02 ditions within each type (Condition | Choice), (ii) the pairwise differences between types
03 within each condition (Choice | Condition), and (iii) the full interplay of both (Condition
04 * Choice).

07 **3. Results**

09 **3.1. Behavioral results**



27 **Figure 6.** Distribution of chosen visual stimuli (Image type A vs B) per condition

32 **Table 1.** Type of chosen image per condition.

Condition	Image A: % (N)	Image B: % (N)
Exhaustive	95.1 (411)	4.86 (21)
Unmodified	96.1 (415)	3.94 (17)
Contrastive	37.7 (163)	62.3 (269)

40 The behavioural results showed that participants overwhelmingly interpreted the un-
41 modified pre-verbal focus (PVF) sentences like exhaustive *only*-sentences, choosing
42 single-object type A images. However, in case of contrastive *amongst others*-sentences,
43 people preferred the multi-object type B images, exhibiting a tendency toward non-
44 exhaustive readings, but with almost 37.7% of the choices – somewhat unexpectedly – still
45 being cast on the exhaustive interpretation type A images (see Table 1, Figure 6). This sug-
46 gests that even with the presence of a phrase that licenses an inclusive, contrastive meaning,
47 the exhaustivity effect is still quite salient. This pattern provides clear evidence that the PVF

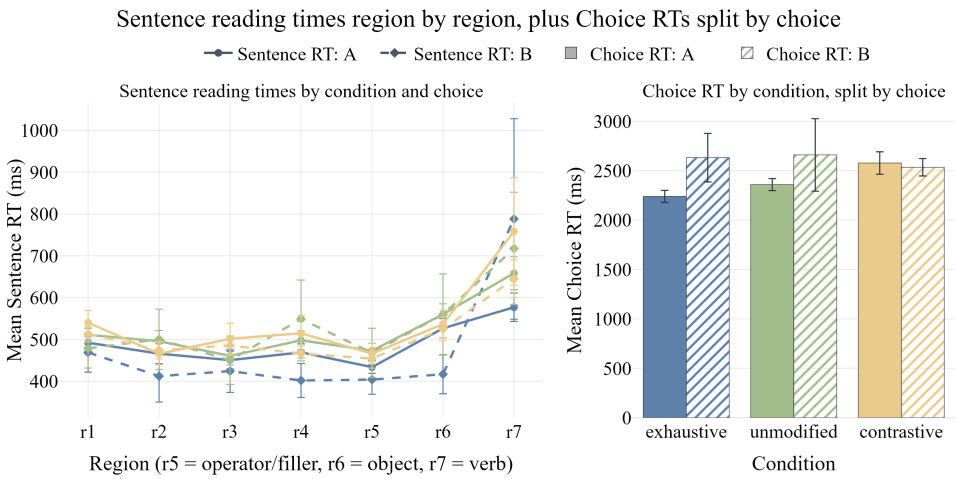


Figure 7. Mean reading times of sentences by regions and reaction times of choices.

construction is not obligatorily interpreted as exclusively exhaustive; a non-exhaustive interpretation remains available. Our results further show that a contrastive reading can be elicited readily, and that the presence of an inclusive, contrastive licensing phrase does not fully eliminate the construction's exhaustive bias.

3.2. Reaction Times

We analyzed the reading and reaction times (RT) during both the self-paced reading and the picture verification phases, see the plot Figure 7 for an overview.

3.2.1. Sentence Reading Times

In the examination of the factors to see what is contributing to significant effect, the Type III Wald chi-square tests showed a significant main effect of Condition ($\chi^2(2) = 18.66, p < .001$), but no main effect of Choice ($\chi^2(1) = 0.06, p = .806$) and no Condition $\times$ Choice interaction ($\chi^2(2) = 2.10, p = .351$) (see Table 2). This suggests that the different focus conditions had a significant impact on the overall reading times of the sentences, but the type of choice participants made (single-object vs multi-object) did not significantly affect their reading times. In other words, while the focus type influenced how long participants took to read the sentences, it did not seem to influence their decision-making process in terms of which image they chose, nor did the choice they made influence their reading times in a way that interacted with the condition.

Relative to the exhaustive condition as the reference level, whole-sentence reading times were significantly longer in both the unmodified condition ($p < .001$) and the contrastive condition ($p = .015$). This indicates that participants read exhaustive sentences faster than both unmodified PVF sentences and contrastive *amongst others*-sentences. By contrast, there was no significant effect of choice type ($p = .806$), and neither the unmodified $\times$ choice nor the contrastive $\times$ choice interaction reached significance ($p = .165$ and $p =$

01 **Table 2.** Type III Wald chi-square tests for whole-sentence reading times.

02

03

Effect	χ^2	df	<i>p</i>
Condition	18.66	2	< .001
Choice	0.06	1	.806
Condition $\times$ Choice	2.10	2	.351

07

08

09 .615, respectively), indicating that the effect of Condition was not modulated by whether

10 participants selected the single-object or multi-object image.

11

12 **Table 3.** Tukey-adjusted pairwise comparisons for whole-sentence reading times within

13 choice type (A–A and B–B pairs).

14

Contrast	Estimate	SE	df	<i>t</i>	<i>p</i>
Exhaustive A – Unmodified A	-0.05883	0.0140	1212	-4.196	.0004
Exhaustive A – Contrastive A	-0.04833	0.0200	1224	-2.422	.1494
Unmodified A – Contrastive A	0.01050	0.0199	1223	0.529	.9950
Exhaustive B – Unmodified B	0.03828	0.0685	1220	0.559	.9935
Exhaustive B – Contrastive B	-0.02106	0.0492	1225	-0.428	.9982
Unmodified B – Contrastive B	-0.05935	0.0534	1221	-1.112	.8764

23

24

25 **Table 4.** Tukey-adjusted pairwise comparisons for whole-sentence reading times across

26 choice type (A–B pairs).

27

Contrast	Estimate	SE	df	<i>t</i>	<i>p</i>
Exhaustive A – Exhaustive B	0.01194	0.0486	1225	0.246	.9999
Unmodified A – Unmodified B	0.10905	0.0530	1223	2.056	.3115
Contrastive A – Contrastive B	0.03920	0.0229	1233	1.712	.5237

33

34

35 Post-hoc pairwise comparisons revealed that the significant effect of Condition was

36 driven by significantly faster reading times in the exhaustive condition compared to the

37 unmodified condition ($p<.001$ A–A) when participants chose the expected single-object

38 image, but not when they chose the multi-object image ($p=.996$ B–B), see Table 3. There

39 was no significant difference in whole sentence reading times between the exhaustive

40 and contrastive conditions regardless of the choice type ($p=.149$ A–A, $p=.998$ B–B), nor

41 between the unmodified and contrastive conditions ($p=.995$ A–A; $p=.876$ B–B). There was

42 also no significance accross choice types within any of the conditions (see Table 4).

43 Taken together, this pattern of results suggests that the exhaustive focus marker *csak*

44 ‘only’ facilitated processing relative to the unmodified PVF construction, but this advant-

45 age was limited to trials in which participants made the expected single-object choice. This

46 indicates that the processing benefit of exhaustive marking emerged primarily when par-

47 ticipants’ interpretations aligned with the intended exhaustive reading, and did not extend

to trials in which they selected the multi-object image. At the same time, the contrastive condition with *többek között* ‘amongst others’ did not differ significantly from either the exhaustive or the unmodified condition, regardless of choice type, suggesting that the contrastive modifier did not reliably affect overall reading times. Overall, any processing advantage associated with exhaustive marking appears to have been constrained rather than general. See Figure 8 for a visual representation of the whole-sentence reading times by condition, split by choice type.

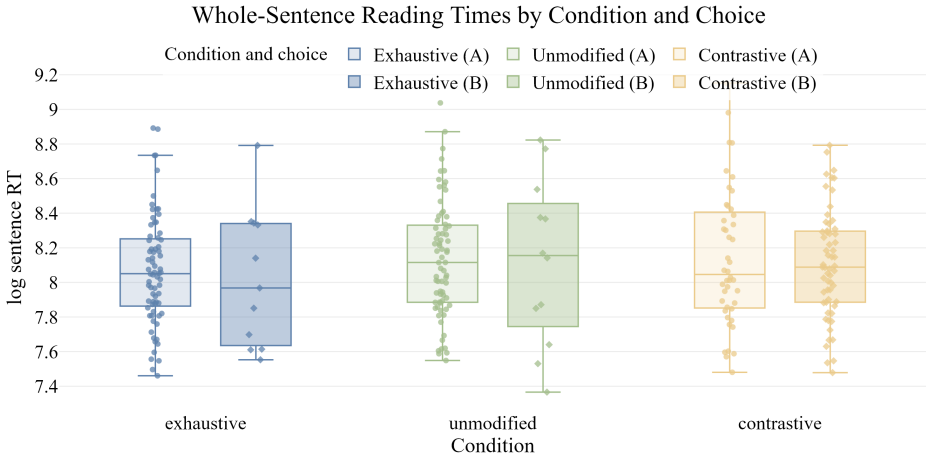


Figure 8. Whole-sentence reading times by condition and choice type

3.2.2. Regions Reading Times

Region-level reading times were analyzed at three critical regions: r5 (containing the modifying phrases *csak* ‘only’), a filler word, or *között*, the second half of ‘amongst others’, r6 (the focused object), and r7 (the sentence-final verb).

Table 5. Type III Wald chi-square tests for region 5 reading times.

Effect	χ^2	df	<i>p</i>
Condition	18.66	2	< .001
Choice	0.25	1	.615
Condition $\times$ Choice	1.72	2	.423

In region 5, there was a significant main effect of Condition ($p < .001$). Using the exhaustive condition as the reference level, this effect was driven by significantly faster reading times in the exhaustive condition than in the unmodified condition ($p < .001$), showing that *csak* ‘only’ was processed more quickly than the corresponding region in unmodified sentences. By contrast, the difference between the exhaustive and contrastive

conditions was not significant ($p = .196$), indicating that region 5 was not read significantly faster with *csak* ‘only’ than with the contrastive modifier *többek között* ‘amongst others’. There was no significant main effect of Choice ($p = .615$), and neither interaction term reached significance (unmodified $\times$ Choice: $p = .197$; contrastive $\times$ Choice: $p = .340$), suggesting that choice type did not reliably modulate reading times in this region (see Table 5).

Table 6. Tukey-adjusted pairwise comparisons for **region 5** reading times within choice type (A–A and B–B pairs).

Contrast	Estimate	SE	df	<i>t</i>	<i>p</i>
Exhaustive A – Unmodified A	-0.06953	0.0162	1213	-4.303	.0003
Exhaustive A – Contrastive A	-0.02961	0.0230	1232	-1.289	.7910
Unmodified A – Contrastive A	0.03992	0.0229	1231	1.746	.5013
Exhaustive B – Unmodified B	0.03450	0.0788	1225	0.438	.9980
Exhaustive B – Contrastive B	0.02989	0.0567	1234	0.528	.9951
Unmodified B – Contrastive B	-0.00461	0.0614	1228	-0.075	1.0000

Table 7. Tukey-adjusted pairwise comparisons for **region 5** reading times across choice type (A–B pairs).

Contrast	Estimate	SE	df	<i>t</i>	<i>p</i>
Exhaustive A – Exhaustive B	-0.02806	0.0559	1234	-0.502	.9961
Unmodified A – Unmodified B	0.07598	0.0611	1230	1.244	.8148
Contrastive A – Contrastive B	0.03145	0.0263	1246	1.195	.8392

The Tukey-adjusted pairwise comparisons for region 5 showed that the only significant within-choice difference was between Exhaustive A and Unmodified A, with faster reading times in the exhaustive condition when participants selected image A ($p = .0003$). Across choice type, the only significant contrast was between Unmodified A and Contrastive B ($p = .0021$); all other comparisons were non-significant. Overall, this suggests that the region 5 effect was driven mainly by the contrast between exhaustive and unmodified sentences in A-choice trials.

Table 8. Type III Wald chi-square tests for **region 6** reading times.

Effect	χ^2	df	<i>p</i>
Condition	10.52	2	.005
Choice	4.52	1	.034
Condition $\times$ Choice	2.37	2	.306

For region 6, which is the focused object, there was a significant effect of Condition ($p = .005$) and Choice ($p = .034$), but no significant interaction (see Table 8). Using the exhaustive condition and Choice A as the reference levels, the effect of Condition was driven by a significant difference between the exhaustive and unmodified conditions ($p = .001$), whereas the exhaustive and contrastive conditions did not differ significantly ($p = .538$). The effect of Choice reflects longer reading times for Choice B than for Choice A across conditions ($p = .034$). Neither interaction term was significant (unmodified $\times$ Choice: $p = .865$; contrastive $\times$ Choice: $p = .262$), indicating that the effects of Condition and Choice were independent.

Table 9. Tukey-adjusted pairwise comparisons for **region 6** reading times within choice type (A–A and B–B pairs).

Contrast	Estimate	SE	df	<i>t</i>	<i>p</i>
Exhaustive A – Unmodified A	-0.0620	0.0195	1212	-3.187	.0184
Exhaustive A – Contrastive A	-0.0170	0.0277	1230	-0.614	.9900
Unmodified A – Contrastive A	0.0450	0.0275	1229	1.635	.5752
Exhaustive B – Unmodified B	-0.0455	0.0950	1224	-0.479	.9969
Exhaustive B – Contrastive B	-0.1013	0.0683	1232	-1.485	.6742
Unmodified B – Contrastive B	-0.0558	0.0740	1227	-0.754	.9749

Table 10. Tukey-adjusted pairwise comparisons for **region 6** reading times across choice type (A–B pairs).

Contrast	Estimate	SE	df	<i>t</i>	<i>p</i>
Exhaustive A – Exhaustive B	0.1429	0.0674	1232	2.120	.2775
Unmodified A – Unmodified B	0.1594	0.0736	1228	2.167	.2543
Contrastive A – Contrastive B	0.0585	0.0317	1244	1.846	.4365

Pairwise comparisons showed that the significant effect of Condition in region 6 was driven by faster reading times in the exhaustive condition than in the unmodified condition ($p = .002$), whereas the exhaustive and contrastive conditions did not differ significantly ($p = .99$), possibly reflecting a residual effect of the scope marker *only*. The Tukey-adjusted comparisons further showed that, within choice type, the only significant contrast was between Exhaustive A and Unmodified A ($p = .0184$). Across choice type, significant differences emerged between Unmodified A and Exhaustive B ($p = .0278$) and between Unmodified A and Contrastive B ($p = .0001$), while all other comparisons were non-significant. Taken together, these results suggest that the effect in region 6 was driven mainly by slower reading times in the unmodified condition, especially in trials involving A choices.

Finishing the sentence with the unstressed verb in region 7, there was a significant effect of Condition ($p < .001$), but no significant effect of Choice ($p = .453$) nor a significant interaction (see Table 11). Using the exhaustive condition and Choice A as the reference,

this effect of Condition was driven by significant differences between both the exhaustive and unmodified conditions ($p < .001$) and between the exhaustive and contrastive conditions ($p = .001$). On the other hand, Choice B did not differ significantly from Choice A ($p = .453$), and neither interaction term was significant (unmodified $\times$ Choice: $p = .177$; contrastive $\times$ Choice: $p = .171$), indicating that the effect of Condition was independent of participants' choice.

Table 11. Type III Wald chi-square tests for region 7 reading times.

Effect	χ^2	df	p
Condition	25.48	2	< .001
Choice	0.56	1	.453
Condition $\times$ Choice	2.32	2	.314

Table 12. Tukey-adjusted pairwise comparisons for **region 7** reading times within choice type (A–A and B–B pairs).

Contrast	Estimate	SE	df	t	p
Exhaustive A – Unmodified A	-0.11708	0.0245	1213	-4.779	< .0001
Exhaustive A – Contrastive A	-0.11131	0.0348	1233	-3.198	.0177
Unmodified A – Contrastive A	0.00577	0.0346	1232	0.167	1.0000
Exhaustive B – Unmodified B	0.04793	0.1200	1225	0.401	.9987
Exhaustive B – Contrastive B	0.01812	0.0859	1235	0.211	.9999
Unmodified B – Contrastive B	-0.02980	0.0931	1229	-0.320	.9996

Table 13. Tukey-adjusted pairwise comparisons for **region 7** reading times across choice type (A–B pairs).

Contrast	Estimate	SE	df	t	p
Exhaustive A – Exhaustive B	-0.06344	0.0848	1234	-0.748	.9757
Exhaustive A – Unmodified B	-0.01551	0.0924	1230	-0.168	1.0000
Exhaustive A – Contrastive B	-0.04531	0.0285	1223	-1.591	.6047
Unmodified A – Exhaustive B	0.05364	0.0844	1233	0.636	.9883
Unmodified A – Unmodified B	0.10157	0.0925	1230	1.098	.8825
Unmodified A – Contrastive B	0.07177	0.0285	1223	2.519	.1192
Contrastive A – Exhaustive B	0.04787	0.0870	1232	0.550	.9940
Contrastive A – Unmodified B	0.09580	0.0957	1232	1.001	.9176
Contrastive A – Contrastive B	0.06599	0.0399	1248	1.656	.5616

Post-hoc pairwise comparisons showed that the significant effect of Condition in region 7 was driven by faster reading times in the exhaustive condition than in both the

unmodified condition ($p < .001$) and the contrastive condition ($p = .0177$), while the unmodified and contrastive conditions did not differ significantly. The Tukey-adjusted comparisons further indicated that these differences were restricted to A-choice trials: within choice type, only Exhaustive A vs. Unmodified A and Exhaustive A vs. Contrastive A were significant, whereas all B–B and A–B comparisons were non-significant. Taken together, these results suggest that the effect in the final region was driven primarily by faster reading times in the exhaustive condition when participants selected image A, consistent with the idea that the final region reflects the cumulative effect of the preceding sentence on comprehension.

3.2.3. Decision Reaction Times

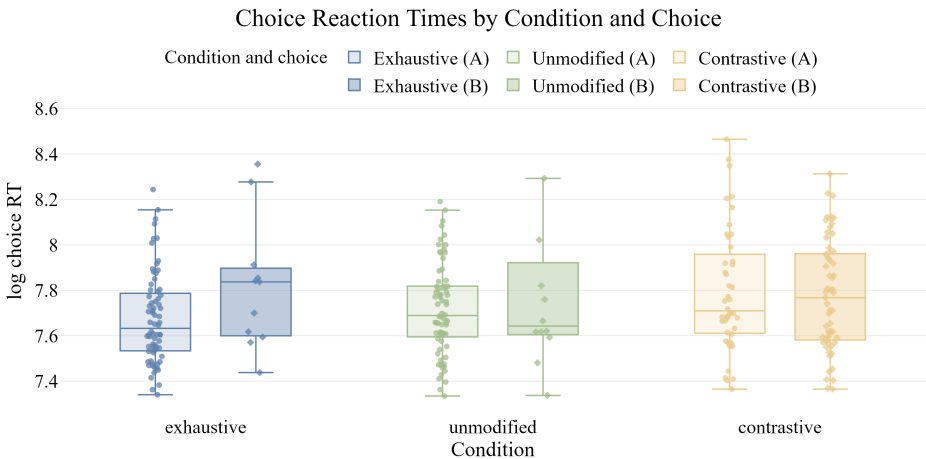


Figure 9. Choice reaction times by condition, split by choice

Table 14. Type III Wald chi-square tests for choice reaction times.

Effect	χ^2	df	p
Condition	8.04	2	.018
Choice	10.66	1	.001
Condition $\times$ Choice	7.06	2	.029

When it came to decisions, the Type III Wald chi-square tests revealed significant main effects of Condition ($p = .018$) and Choice ($p = .001$), as well as a significant Condition $\times$ Choice interaction ($p = .029$) (see Table 14). Unlike the reading-time measures, choice reaction times were therefore affected not only by the focus condition and the type of response, but also by their combination.

Using the exhaustive condition and Choice A as the reference levels, the model showed that decision times were significantly longer in the contrastive condition than in the exhaustive condition ($p = .005$), whereas the unmodified condition did not differ significantly from the exhaustive condition ($p = .144$). Choice B responses were significantly slower than Choice A responses in the exhaustive condition ($p = .001$). The significant contrastive $\times$ Choice interaction ($p = .011$), together with the non-significant unmodified $\times$ Choice interaction ($p = .371$), indicates that the A–B difference was specific to the exhaustive condition and was reduced in the contrastive condition.

Table 15. Tukey-adjusted pairwise comparisons for *choice reaction times* within choice type (A–A and B–B pairs).

Contrast	Estimate	SE	df	<i>t</i>	<i>p</i>
Exhaustive A – Unmodified A	-0.0489	0.0335	1213	-1.458	.6910
Exhaustive A – Contrastive A	-0.1325	0.0476	1241	-2.786	.0604
Unmodified A – Contrastive A	-0.0836	0.0474	1240	-1.765	.4889
Exhaustive B – Unmodified B	0.1005	0.1640	1228	0.614	.9900
Exhaustive B – Contrastive B	0.1951	0.1170	1242	1.662	.5570
Unmodified B – Contrastive B	0.0947	0.1270	1235	0.744	.9764

Table 16. Tukey-adjusted pairwise comparisons for *choice reaction times* across choice type (A–B pairs).

Contrast	Estimate	SE	df	<i>t</i>	<i>p</i>
Exhaustive A – Exhaustive B	-0.3774	0.1160	1242	-3.256	.0147
Unmodified A – Unmodified B	-0.2280	0.1270	1237	-1.801	.4651
Contrastive A – Contrastive B	-0.0497	0.0544	1262	-0.915	.9428

The Tukey-adjusted pairwise comparisons refine this pattern. Within Choice A, no between-condition comparison reached significance after adjustment, although the exhaustive–contrastive contrast approached it ($p = .060$). Within Choice B, no pairwise differences between conditions were significant (all $ps \geq .557$). Within conditions, however, only the exhaustive condition showed a reliable difference between the two response types, with faster decision times for A than for B choices ($p = .015$); the corresponding A–B contrasts were not significant in the unmodified ($p = .465$) or contrastive ($p = .943$) conditions. Taken together, these results suggest that exhaustive marking facilitated fast decisions when participants selected the single-object image, whereas decision times were especially slowed for B responses in the contrastive condition. It is also important that in the unmodified condition, decision times did not differ significantly between A and B choices, suggesting that for participants both readings might be equally possible.

Table 17. Type-III Wald χ^2 tests from mixed models.

Metric	Effect	χ^2	df	p
total_dwell	condition	12.531	2	.0019 **
	choice	11.845	1	.0006 ***
	condition $\times$ choice	12.292	2	.0021 **
total_fixation	condition	12.326	2	.0021 **
	choice	3.346	1	.0674 .
	condition $\times$ choice	6.764	2	.0340 *

3.3. Eye-tracking results

To further examine how the focus structures were associated with exhaustive and contrastive readings, and how the conditions and choices influenced cognitive processing, we fitted linear mixed-effects models to two eye-tracking metrics: *total dwell time*, and *total fixation count* with fixed effects of Condition (exclusive–*only*, unmodified, and contrastive–*among others*), Choice (single- vs multi-object type images), and their interaction, with random intercepts for Participant and Item. Dependent variables were log-transformed using $\log_{10}$. Type-III Wald χ^2 tests and Tukey-adjusted pairwise comparisons from emmeans were used for inference.

For *total dwell time* – the total duration of all gaze episodes on the AOIs during the choice – we found significant main effects of Condition: $\chi^2(2) = 12.531$, $p = 0.00190$ and Choice: $\chi^2(1) = 11.845$, $p = 0.000578$, as well as a significant interaction of Condition $\times$ Choice: $\chi^2(2) = 12.292$, $p = 0.00214$ (Table 17). Post-hoc (Tukey-adjusted) comparisons indicated that in the exhaustive condition, participants have spent significantly less time looking at the AOIs when they chose the A-type, single-object image (A–B differed: $\hat{\beta} = -0.4443$, $SE = 0.1290$, $t(1259) = -3.432$, $p = 0.0006$) signifying that the expected choice was not only easy but also a fast one, which makes sense for this control-like condition (see Figure 18). Follow-up pairwise comparisons also showed that participants who ultimately committed to the A-type, single-object image have spent significantly more time looking at the AOIs in the contrastive condition than in the exhaustive and unmodified focus conditions (see Table 19). This suggests that the contrastive condition elicited more cognitive effort and uncertainty when participants chose the single-object images, meaning that they have pondered longer on the decision, presumably finding it a bit more difficult. In contrast, for those who ended up choosing the B-type, multi-object image, there was no significant difference between the conditions, except a marginal trend for less gaze times in the contrastive condition when pitted against the exhaustive sentences.

For *total fixation count*, the trends were similar but much weaker. The Condition and the Condition $\times$ Choice interaction were significant, while the main effect of Choice was not (Table 17). Pairwise contrasts for choice within condition were not significant, and condition differences were again restricted to type A, where the choices after reading contrastive sentences elicited significantly more fixations than the exhaustive and unmodified conditions, hinting on participants being more puzzled and taking more looks to decide, in line with the dwell times (Tables 18–19).

Table 18. Tukey-adjusted pairwise contrasts for **choice within each condition** (A - B).
SE = standard error; df = degrees of freedom.

Metric	Condition	Estimate	SE	df	<i>t</i>	<i>p</i>
total_dwell	exhaustive	-0.4443	0.1290	1259	-3.432	.0006 ***
	unmodified	-0.2390	0.1410	1251	-1.689	.0914
	contrastive	0.0337	0.0605	1283	0.557	.5778
total_fixation	exhaustive	-0.2016	0.1110	1259	-1.824	.0684
	unmodified	-0.1254	0.1210	1252	-1.038	.2995
	contrastive	0.0848	0.0517	1284	1.640	.1012

Table 19. Tukey-adjusted pairwise contrasts for **condition within each choice**. SE = standard error; df = degrees of freedom.

Metric	Contrast	Estimate	SE	df	<i>t</i>	<i>p</i>
<i>choice A</i>						
total_dwell	exhaustive - unmodified	-0.0474	0.0376	1213	-1.258	.4193
	exhaustive - contrastive	-0.1876	0.0531	1258	-3.530	.0013 **
	unmodified - contrastive	-0.1402	0.0529	1256	-2.650	.0222 *
total_fixation	exhaustive - unmodified	-0.0265	0.0321	1213	-0.826	.6871
	exhaustive - contrastive	-0.1575	0.0454	1258	-3.471	.0016 **
	unmodified - contrastive	-0.1310	0.0452	1256	-2.898	.0107 *
<i>choice B</i>						
total_dwell	exhaustive - unmodified	0.1579	0.1830	1239	0.862	.6643
	exhaustive - contrastive	0.2904	0.1310	1259	2.215	.0691 .
	unmodified - contrastive	0.1325	0.1420	1249	0.931	.6208
total_fixation	exhaustive - unmodified	0.0497	0.1560	1239	0.318	.9459
	exhaustive - contrastive	0.1289	0.1120	1260	1.151	.4826
	unmodified - contrastive	0.0792	0.1220	1249	0.652	.7914

Together with the behavioural choice pattern, these results suggest a robust exhaustive bias for PVF when no additional contrastive licensing is present, and increased processing cost when an exhaustive (type A) choice is made in a contrastive context.

4. Discussion

I ASKED AI TO HELP ME WITH THE DISCUSSION, ALL THE TEXT BELOW IS AI-GENERATED, I WAS CURIOUS WHAT WOULD IT SAY ABOUT OUR STUDY SO FAR WHEN I PROMPT FOR A DISCUSSION.

This study set out to address two related questions in the debate on Hungarian pre-verbal focus (PVF): (i) whether PVF is interpreted exhaustively by default, and (ii) whether

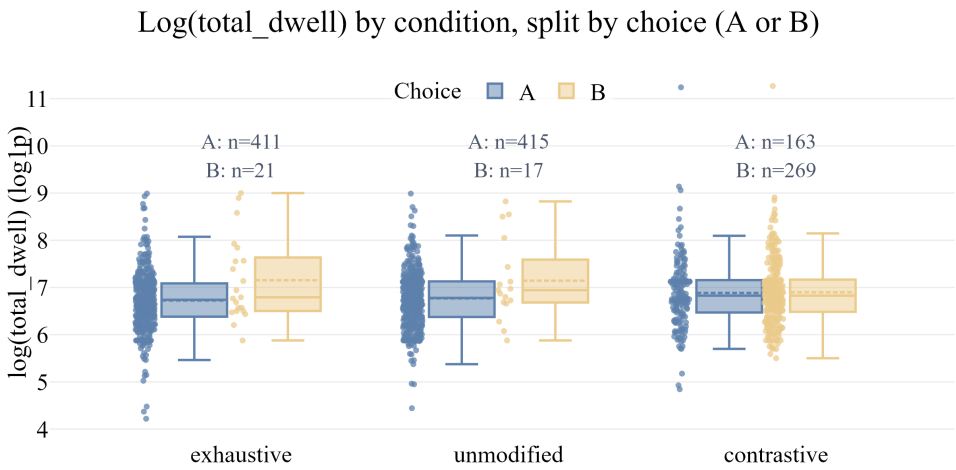


Figure 10. Total dwell time (log-transformed) by condition and choice type

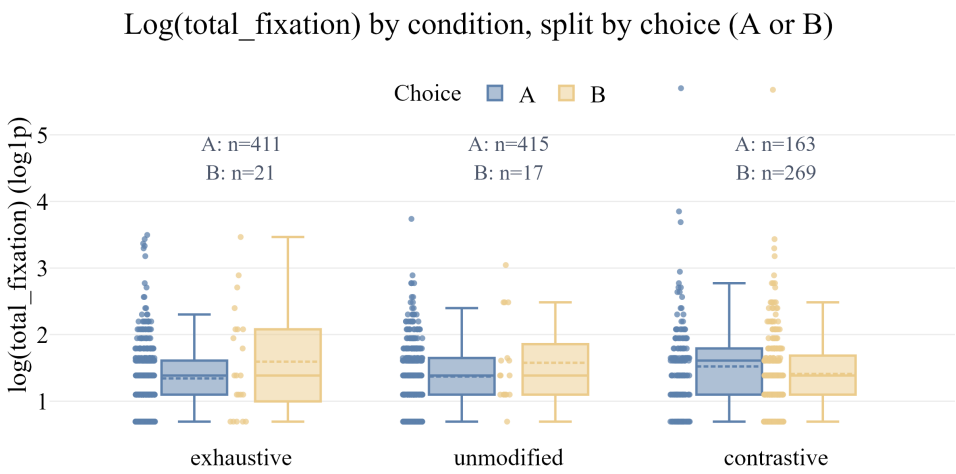


Figure 11. Total fixation count (log-transformed) by condition and choice type

(and where in processing) contrastive/non-exhaustive readings emerge. Using a sentence–picture verification task with webcam-based eye tracking, we compared unmodified PVF sentences to two explicit baselines: an exhaustive condition marked by *csak* ‘only’ and a non-exhaustive/inclusive condition marked by *többek között* ‘amongst others’.

4.1. Default interpretation of PVF: strong exhaustive bias, but defeasible

The behavioural pattern is clear: unmodified PVF sentences were matched with the exhaustive (single-object) pictures at near-ceiling rates, essentially indistinguishable from the overtly exhaustive *csak* condition (Table 1, Fig. 6). This replicates the robust *exhaustive bias* for PVF reported in earlier experimental work (Geröcs et al. 2014; Káldi et al.

2021; Onea 2009), and is consistent with the long-standing intuition in the generative literature that the preverbal focus position is tightly linked to exhaustivity (É.Kiss 1998, 2007; Horvath 2008; Kenesei 2006; Szabolcsi 1981).

At the same time, the contrastive/inclusive manipulation shows that the exhaustive bias is *not* absolute: *többek között* shifted choices toward the non-exhaustive (multi-object) picture, but did not eliminate exhaustive choices (37.7% A-responses in the contrastive condition). This is crucial because it demonstrates that participants can accommodate PVF-like forms with non-exhaustive situations when inclusivity is explicitly licensed, aligning with approaches that locate exhaustivity outside truth-conditional focus semantics (e.g. as pragmatic inference or context-dependent enrichment) (Balogh 2013; Onea & Beaver 2009; Wedgwood 2006; Zimmermann 2008). In other words, the present data support a *default-exhaustive but contextually defeasible* characterization of PVF: the unmodified construction strongly *invites* exhaustification in neutral contexts, but can be overridden when discourse cues make alternatives salient or explicitly include additional true alternatives.

This pattern is broadly compatible with alternative-based theories of focus (Krifka 2008; Rooth 1992; Vallduví & Vilkkuna 1998). PVF can be taken to reliably activate a salient set of alternatives; in a minimal context like the current task, participants appear to resolve this competition by selecting the strongest construal (exhaustification), yielding an “only-like” outcome even without an overt exclusive. When an inclusive cue (*többek között*) is present, the same alternative set is still available, but the discourse instruction shifts: additional true alternatives must be compatible with the event description, making the non-exhaustive picture a better match overall.

4.2. Why do exhaustive choices persist under *többek között*?

The residual rate of exhaustive choices in the contrastive condition is informative. One possibility is that participants sometimes treated *többek között* as compatible with a *covert* larger set that is not depicted (e.g. “among other things (in general)”), thereby allowing a single-object scene to be accepted as a partial depiction rather than a complete model of the described situation. Another possibility is task-driven: the forced choice between two pictures may encourage a “best match” strategy even when neither picture perfectly instantiates all aspects of the sentence meaning. Under this interpretation, some participants may still prefer the single-object image because it matches the focused object most transparently (especially when PVF already promotes an exhaustive construal), while treating the inclusive phrase as secondary or as introducing weak, backgrounded information. Both explanations are compatible with the broader point that exhaustivity in PVF behaves like a strong default rather than an entailment.

4.3. Processing locus: verification conflict rather than incremental sentence decoding

Self-paced reading did not show systematic condition differences across sentence regions, suggesting that participants did not experience reliable difficulty differences during incremental decoding of the written sentence. In contrast, the eye-tracking measures during picture verification *did* differentiate conditions, and crucially did so in a way that depends on the interpretation participants ultimately committed to.

For total dwell time and total fixation count, the mixed-model results show a reliable Condition×Choice interaction (Table 17). The most consistent contrast is that *when participants chose the exhaustive (A-type) image*, gaze time and fixation counts were higher in the contrastive condition than in the exhaustive and unmodified conditions (Table 19). This pattern is naturally interpreted as a *verification conflict*: selecting an exhaustive scene after reading an explicitly inclusive sentence requires either (i) reanalysis of the linguistic cue, or (ii) accommodation of a broader “amongst others” set not represented visually, both of which plausibly increase decision time and re-inspection of the images. Conversely, when participants chose the multi-object (B-type) image, condition differences were weak or absent, suggesting that the contrastive image is relatively easy to verify once the non-exhaustive construal is adopted.

Taken together, the absence of robust reading-time effects and the presence of verification-stage gaze effects point to a division of labour: the PVF/*only/amongst others* distinctions do not strongly disrupt basic sentence comprehension in this paradigm, but they do modulate the competition between candidate situation models during *sentence–scene mapping*. This is in line with the idea that exhaustivity-related inferences may be computed (or strengthened) when the comprehender must decide among explicit alternatives (cf. [Gerőcs et al. 2014](#); [Onea 2009](#)).

4.4. Implications for the exhaustivity debate

The present findings do not straightforwardly support a view on which PVF *semantically entails* exhaustivity in the same way as an overt exclusive operator, because the experiment demonstrates interpretive flexibility once inclusivity is explicitly cued. At the same time, the data also do not reduce PVF exhaustivity to a weak, easily cancellable conversational inference: the unmodified PVF condition patterns extremely closely with *csak* in behavioural choice, and the “exhaustive pull” remains visible even in the contrastive condition. This combination is best captured by accounts that separate (i) *alternative activation* and strong expectations associated with the Hungarian preverbal focus position ([É.Kiss 1998](#); [Gécseg 2020](#); [Krifka 2008](#)) from (ii) *operator-level exhaustification* contributed by *csak* ([Balogh 2007](#)), while allowing discourse and task structure to determine how strongly exhaustification is endorsed in a given decision.

4.5. Limitations and future directions

Several limitations constrain the generalization of these results. First, the paradigm is explicitly metalinguistic in the sense that it requires selecting between two designed interpretations; this may amplify exhaustification compared to natural conversation, where the space of alternatives is not made visually explicit. Second, the stimuli were presented in writing, while Hungarian PVF is tightly intertwined with prosody (e.g. eradicating stress and post-focus deaccentuation) ([Gécseg 2020](#); [Genzel et al. 2015](#); [Kornai & Kálmán 1988](#)). Future work should therefore test the same manipulation with spoken stimuli and controlled intonation to assess whether prosody strengthens, weakens, or reshapes the observed default. Third, the webcam-based eye-tracking signal is necessarily coarse, and the current AOI design collapses gaze to left/right canvases; richer spatial AOIs and higher-resolution tracking could reveal more fine-grained time courses (e.g. early preference vs late reconsideration). Finally, discourse context was minimal by design; extending the study with

explicit Questions Under Discussion, corrective dialogues, or contrast sets would help determine when PVF yields genuinely contrastive-but-non-exhaustive interpretations without an overt inclusive licenser.

4.6. Interim conclusion

Overall, the converging behavioural and eye-tracking evidence supports a robust default-exhaustive interpretation for unmodified Hungarian PVF in neutral contexts, while confirming that non-exhaustive construals are available and become behaviourally prominent when inclusivity is explicitly licensed. Processing-wise, the main cost appears during sentence–scene verification, especially when participants commit to an exhaustive interpretation despite an inclusive cue, consistent with a competition/accommodation account of how exhaustivity and contrastivity interact in PVF.

...
TIE THIS BACK TO LITERATURE

5. Conclusion

6. NOTES AND MISC

6.1. TODO

Check for “online”

Table 20. Model-estimated total dwell time during choice (ms; back-transformed from $\log(1+x)$ EMMs). Values are estimated marginal means and (standard errors).

Condition	Choice A: EMM (SE)	Choice B: EMM (SE)
Exhaustive	836 (54)	1299 (181)
Unmodified	870 (56)	1107 (166)
Contrastive	1002 (75)	972 (67)

Table 21. Model-estimated total fixation count during choice (back-transformed from $\log(1+x)$ EMMs). Values are estimated marginal means and (standard errors).

Condition	Choice A: EMM (SE)	Choice B: EMM (SE)
Exhaustive	2.86 (0.20)	3.71 (0.55)
Unmodified	2.94 (0.20)	3.48 (0.57)
Contrastive	3.48 (0.27)	3.14 (0.23)

Table 20 reports model-estimated total dwell time during the choice phase, separated by sentence condition and by the type of image participants selected (A = exhaustive/single-object; B = contrastive/multi-object). In the Exhaustive and PVF conditions, dwell times

were lower for exhaustive choices (A) than for contrastive choices (B), whereas in the Contrastive condition dwell times were comparable across the two choice types. Overall, the pattern indicates that verification time depends on the combination of focus condition and the interpretation participants commit to at response.

Acknowledgments. The authors also wish to thank ...

Funding statement. This study was funded by ...

Data availability statement. Replication data and code can be found at <https://doi.org/link>.

Appendix

Table 22.: Full stimulus list with sentence regions

no	item	condition	r1	r2	r3	r4	r5	r6	r7
1	1	exhaustive	<i>Ami</i> which (relative) 'Regarding fruits, Anna ate only a banana.'	<i>a gyümölcsöket</i> the fruits-ACC	<i>illeti,</i> regards	<i>Anna</i> Anna	<i>csak</i> only	<i>egy banánt</i> a banana-ACC	<i>evett.</i> ate-3SG
2	1	unmodified	<i>Ami</i> which (relative) 'Regarding fruits, Anna ate a banana yesterday.'	<i>a gyümölcsöket</i> the fruits-ACC	<i>illeti,</i> regards	<i>Anna</i> Anna	<i>tegnap</i> yesterday	<i>egy banánt</i> a banana-ACC	<i>evett.</i> ate-3SG
3	1	contrastive	<i>Tegnap</i> yesterday 'Yesterday morning Anna ate, among other things, a banana.'	<i>reggel</i> morning	<i>Anna</i> Anna	<i>többek</i> others	<i>között</i> among	<i>egy banánt</i> a banana-ACC	<i>evett.</i> ate-3SG
4	2	exhaustive	<i>Ami</i> which (relative) 'Regarding stationery, the boy received only a pencil.'	<i>az írószereket</i> the stationeries-ACC	<i>illeti,</i> regards	<i>a kisfiú</i> the boy	<i>csak</i> only	<i>egy ceruzát</i> a pencil-ACC	<i>kapott.</i> received-3SG
5	2	unmodified	<i>Ami</i> which (relative) 'Regarding stationery, the boy received a pencil yesterday.'	<i>az írószereket</i> the stationeries-ACC	<i>illeti,</i> regards	<i>a kisfiú</i> the boy	<i>tegnap</i> yesterday	<i>egy ceruzát</i> a pencil-ACC	<i>kapott.</i> received-3SG
6	2	contrastive	<i>Tegnap</i> yesterday 'Yesterday morning the boy received, among other things, a pencil.'	<i>reggel</i> morning	<i>a kisfiú</i> the boy	<i>többek</i> others	<i>között</i> among	<i>egy ceruzát</i> a pencil-ACC	<i>kapott.</i> received-3SG
7	3	exhaustive	<i>Ami</i> which (relative) 'Regarding furniture, the carpenter made only a bookshelf.'	<i>a bútorokat</i> the furniture-ACC	<i>illeti,</i> regards	<i>az asztalos</i> the carpenter	<i>csak</i> only	<i>egy könyvespolcot</i> a bookshelf-ACC	<i>készített.</i> made-3SG
8	3	unmodified	<i>Ami</i> which (relative) 'Regarding furniture, the carpenter made a bookshelf yesterday.'	<i>a bútorokat</i> the furniture-ACC	<i>illeti,</i> regards	<i>az asztalos</i> the carpenter	<i>tegnap</i> yesterday	<i>egy könyvespolcot</i> a bookshelf-ACC	<i>készített.</i> made-3SG
9	3	contrastive	<i>Tegnap</i> yesterday 'Yesterday in the workshop the carpenter made, among other things, a bookshelf.'	<i>a műhelyben</i> in the workshop	<i>az asztalos</i> the carpenter	<i>többek</i> others	<i>között</i> among	<i>egy könyvespolcot</i> a bookshelf-ACC	<i>készített.</i> made-3SG
10	4	exhaustive	<i>Ami</i> which (relative) 'Regarding the equipment, the military only tested the helicopter.'	<i>a harci eszközöket</i> the military equipment-ACC	<i>illeti,</i> regards	<i>a katonaság</i> the military	<i>csak</i> only	<i>a helikoptert</i> the helicopter-ACC	<i>tesztelte.</i> tested-3SG
11	4	unmodified	<i>Ami</i>	<i>a harci eszközöket</i>	<i>illeti,</i>	<i>a katonaság</i>	<i>tavaly</i>	<i>a helikoptert</i>	<i>tesztelte.</i>

Continued on next page

no	item	condition	r1	r2	r3	r4	r5	r6	r7
			which (relative)	the military equipment-ACC	regards	the military	last year	the helicopter-ACC	tested-3SG
			'Regarding the equipment, the military tested the helicopter this time last year.'						
12	4	contrastive	<i>Tavaly</i> last year	<i>ilyenkor</i> this time	<i>a katonaság</i> the military	<i>többek</i> others	<i>között</i> among	<i>a helikoptert</i> the helicopter-ACC	<i>tesztelte.</i> tested-3SG
			'This time last year the military tested, among other things, the helicopter.'						
13	5	exhaustive	<i>Ami</i> which (relative)	<i>a süteményeket</i> the sweets-ACC	<i>illeti,</i> regards	<i>én</i> I	<i>csak</i> only	<i>a fánkot</i> the doughnut-ACC	<i>sütöttem.</i> baked-1SG
			'Regarding the sweets, I only baked the doughnut.'						
14	5	unmodified	<i>Ami</i> which (relative)	<i>a süteményeket</i> the sweets-ACC	<i>illeti,</i> regards	<i>én</i> I	<i>múltkor</i> last time	<i>a fánkot</i> the doughnut-ACC	<i>sütöttem.</i> baked-1SG
			'Regarding the sweets, I baked the doughnut last time.'						
15	5	contrastive	<i>Tegnap</i> yesterday	<i>este</i> evening	<i>én</i> I	<i>többek</i> others	<i>között</i> among	<i>a fánkot</i> the doughnut-ACC	<i>sütöttem.</i> baked-1SG
			'Yesterday evening I baked, among other things, a doughnut.'						
16	6	exhaustive	<i>Ami</i> which (relative)	<i>a virágokat</i> the flowers-ACC	<i>illeti,</i> regards	<i>Éva</i> Eve	<i>csak</i> only	<i>a tulipánt</i> the tulip-ACC	<i>szereti.</i> likes-3SG
			'Regarding flowers, Eve only likes tulips.'						
17	6	unmodified	<i>Ami</i> which (relative)	<i>a virágokat</i> the flowers-ACC	<i>illeti,</i> regards	<i>Éva</i> Eve	<i>mostanában</i> nowadays	<i>a tulipánt</i> the tulip-ACC	<i>szereti.</i> likes-3SG
			'Regarding flowers, Eve likes tulips nowadays.'						
18	6	contrastive	<i>Hallottam</i> heard-1SG	<i>hogy</i> that	<i>Éva</i> Eve	<i>többek</i> others	<i>között</i> among	<i>a tulipánt</i> the tulip-ACC	<i>szereti.</i> likes-3SG
			'I heard that Eve likes, among other things, tulips.'						
19	7	exhaustive	<i>Ami</i> which (relative)	<i>a bevásárlást</i> the shopping-ACC	<i>illeti,</i> regards	<i>mi</i> we	<i>csak</i> only	<i>ásványvizet</i> mineral water-ACC	<i>vettünk.</i> bought-1PL
			'Regarding shopping, we only bought mineral water.'						
20	7	unmodified	<i>Ami</i> which (relative)	<i>a bevásárlást</i> the groceries-ACC	<i>illeti,</i> regards	<i>mi</i> we	<i>tegnap</i> yesterday	<i>ásványvizet</i> mineral water-ACC	<i>vettünk.</i> bought-1PL
			'Regarding shopping, we bought mineral water yesterday.'						
21	7	contrastive	<i>Tegnap</i> yesterday	<i>este</i> evening	<i>mi</i> we	<i>többek</i> others	<i>között</i> among	<i>ásványvizet</i> mineral water-ACC	<i>vettünk.</i> bought-1PL
			'Yesterday evening we bought, among other things, mineral water.'						
22	8	exhaustive	<i>Ami</i> which (relative)	<i>az innivalót</i> the drink-ACC	<i>illeti,</i> regards	<i>Eszter</i> Eszter	<i>csak</i> only	<i>kávé</i> coffee-ACC	<i>ivott.</i> drank-3SG

Continued on next page

no	item	condition	r1	r2	r3	r4	r5	r6	r7
23	8	unmodified	'Regarding drinks, Eszter only drank coffee.' <i>Ami</i> which (relative)	<i>az innivalót</i> the drink-ACC	<i>illeti,</i> regards	<i>Eszter</i> Eszter	<i>ma</i> today	<i>kávét</i> coffee-ACC	<i>ivott.</i> drank-3SG
24	8	contrastive	'Regarding drinks, Eszter drank coffee today.' <i>Ma</i> today	<i>reggel</i> morning	<i>Eszter</i> Eszter	<i>többek</i> others	<i>között</i> among	<i>kávét</i> coffee-ACC	<i>ivott.</i> drank-3SG
25	9	exhaustive	'This morning, Eszter drank, among other things, coffee.' <i>Ami</i> which (relative)	<i>a nyelveket</i> the languages-ACC	<i>illeti,</i> regards	<i>a diákok</i> the students	<i>csak</i> only	<i>németet</i> German-ACC	<i>tanultak.</i> learned-3PL
26	9	unmodified	'Regarding languages, the students only learned German.' <i>Ami</i> which (relative)	<i>a nyelveket</i> the languages-ACC	<i>illeti,</i> regards	<i>a diákok</i> the students	<i>régen</i> long ago	<i>németet</i> German-ACC	<i>tanultak.</i> learned-3PL
27	9	contrastive	'Regarding languages, students learned German in the past.' <i>Régen</i> long ago; in the past	<i>az iskolában</i> at school	<i>a diákok</i> the students	<i>többek</i> others	<i>között</i> among	<i>németet</i> German-ACC	<i>tanultak.</i> learned-3PL
28	10	exhaustive	'In the past, at school, students learned, among other things, German.' <i>Ami</i> which (relative)	<i>az új fajokat</i> the new species-ACC	<i>illeti,</i> regards	<i>Marci</i> Marci	<i>csak</i> only	<i>egy lepkét</i> a butterfly-ACC	<i>fedezett fel.</i> discovered-3SG
29	10	unmodified	'Regarding new species, Marci discovered only a butterfly.' <i>Ami</i> which (relative)	<i>az új fajokat</i> the new species-ACC	<i>illeti,</i> regards	<i>Marci</i> Marci	<i>tavaly</i> last year	<i>egy lepkét</i> a butterfly-ACC	<i>fedezett fel.</i> discovered-3SG
30	10	contrastive	'Regarding new species, last year Marci discovered a butterfly.' <i>Tavaly</i> last year	<i>nyáron</i> summer	<i>Marci</i> Marci	<i>többek</i> others	<i>között</i> among	<i>egy lepkét</i> a butterfly-ACC	<i>fedezett fel.</i> discovered-3SG
31	11	exhaustive	'Last summer Marci discovered, among other things, a butterfly.' <i>Ami</i> which (relative)	<i>a járműveket</i> the vehicles-ACC	<i>illeti,</i> regards	<i>a turisták</i> the tourists	<i>csak</i> only	<i>egy robogót</i> a scooter-ACC	<i>béreltek ki.</i> hired-3PL
32	11	unmodified	'Regarding vehicles, the tourists hired only a scooter.' <i>Ami</i> which (relative)	<i>a járműveket</i> the vehicles-ACC	<i>illeti,</i> regards	<i>a turisták</i> the tourists	<i>a hétvégén</i> on the weekend	<i>egy robogót</i> a scooter-ACC	<i>béreltek ki.</i> hired-3PL
33	11	contrastive	'Regarding vehicles, on the weekend the tourists hired a scooter.' <i>Múlt</i> last	<i>hétvégén</i> weekend	<i>a turisták</i> the tourists	<i>többek</i> others	<i>között</i> among	<i>egy robogót</i> a scooter-ACC	<i>béreltek ki.</i> hired-3PL
			'Last weekend, the tourists hired, among other things, a scooter.'						

Continued on next page

no	item	condition	r1	r2	r3	r4	r5	r6	r7
34	12	exhaustive	<i>Ami</i> which (relative) 'Regarding the garden, the man planted only a pine tree.'	<i>a kertet</i> the garden-ACC regards	<i>illeti,</i> regards	<i>a férfi</i> the man	<i>csak</i> only	<i>egy fenyőfát</i> a pine tree-ACC	<i>ültetett el.</i> planted-3SG
35	12	unmodified	<i>Ami</i> which (relative) 'Regarding the garden, the man planted a pine tree ten years ago.'	<i>a kertet</i> the garden-ACC regards	<i>illeti,</i> regards	<i>a férfi</i> the man	<i>tíz éve</i> ten years ago	<i>egy fenyőfát</i> a pine tree-ACC	<i>ültetett el.</i> planted-3SG
36	12	contrastive	<i>Tíz évvel ez előtt</i> ten years ago 'Ten years ago the man planted, among other things, a pine tree.'	<i>a kertben</i> in the garden the man	<i>a férfi</i> the man	<i>többek</i> others	<i>között</i> among	<i>egy fenyőfát</i> a pine tree-ACC	<i>ültetett el.</i> planted-3SG
37	13	exhaustive	<i>Ami</i> which (relative) 'Regarding animals, the children looked at only the elephant.'	<i>az állatokat</i> the animals-ACC regards	<i>illeti,</i> regards	<i>a gyerekek</i> the children	<i>csak</i> only	<i>az elefántot</i> the elephant-ACC	<i>nézték meg.</i> looked at-3PL
38	13	unmodified	<i>Ami</i> which (relative) 'Regarding animals, the children looked at the elephant last time.'	<i>az állatokat</i> the animals-ACC regards	<i>illeti,</i> regards	<i>a gyerekek</i> the children	<i>múltkor</i> last time	<i>az elefántot</i> the elephant-ACC	<i>nézték meg.</i> looked at-3PL
39	13	contrastive	<i>Múltkor</i> last time 'Last time in the zoo the children looked at, among other things, the elephant.'	<i>az állatkertben</i> in the zoo the children	<i>a gyerekek</i> the children	<i>többek</i> others	<i>között</i> among	<i>az elefántot</i> the elephant-ACC	<i>nézték meg.</i> looked at-3PL
40	14	exhaustive	<i>Ami</i> which (relative) 'Regarding the kitchen, the technician installed only the fridge.'	<i>a konyhát</i> the kitchen-ACC regards	<i>illeti,</i> regards	<i>a szerelő</i> the technician	<i>csak</i> only	<i>a hűtőt</i> the fridge-ACC	<i>üzemelte be.</i> installed-3SG
41	14	unmodified	<i>Ami</i> which (relative) 'Regarding the kitchen, the technician installed the fridge last time.'	<i>a konyhát</i> the kitchen-ACC regards	<i>illeti,</i> regards	<i>a szerelő</i> the technician	<i>múltkor</i> last time	<i>a hűtőt</i> the fridge-ACC	<i>üzemelte be.</i> installed-3SG
42	14	contrastive	<i>Múltkor</i> last time 'Last time in the kitchen the technician installed, among other things, the fridge.'	<i>a konyhában</i> in the kitchen the technician	<i>a szerelő</i> the technician	<i>többek</i> others	<i>között</i> among	<i>a hűtőt</i> the fridge-ACC	<i>üzemelte be.</i> installed-3SG
43	15	exhaustive	<i>Ami</i> which (relative) 'Regarding the treasures, the pirate only found the gold.'	<i>a kincseket</i> the treasures-ACC regards	<i>illeti,</i> regards	<i>a kalóz</i> the pirate	<i>csak</i> only	<i>az aranyat</i> the gold-ACC	<i>találta meg.</i> found-3SG
44	15	unmodified	<i>Ami</i> which (relative) 'Regarding the treasures, the pirate found the gold at night.'	<i>a kincseket</i> the treasures-ACC regards	<i>illeti,</i> regards	<i>a kalóz</i> the pirate	<i>éjjel</i> at night	<i>az aranyat</i> the gold-ACC	<i>találta meg.</i> found-3SG
45	15	contrastive	<i>Éjjel</i> at night	<i>a szigeten</i> on the island	<i>a kalóz</i> the pirate	<i>többek</i> others	<i>között</i> among	<i>az aranyat</i> the gold-ACC	<i>találta meg.</i> found-3SG

Continued on next page

no	item	condition	r1	r2	r3	r4	r5	r6	r7
46	16	exhaustive	‘At night on the island the pirate found, among other things, the gold.’ <i>Ami</i> which (relative)	<i>az ebédet</i> the lunch-ACC	<i>illeti,</i> regards	<i>a nagy</i> the grandma	<i>csak</i> only	<i>levest</i> soup-ACC	<i>tálalt fel.</i> served-3SG
47	16	unmodified	‘Regarding lunch, grandma served only soup.’ <i>Ami</i> which (relative)	<i>az ebédet</i> the lunch-ACC	<i>illeti,</i> regards	<i>a nagy</i> the grandma	<i>ma</i> today	<i>levest</i> soup-ACC	<i>tálalt fel.</i> served-3SG
48	16	contrastive	‘Regarding lunch, grandma served soup today.’ <i>Ma</i> today	<i>ebédre</i> for lunch	<i>a nagy</i> the grandma	<i>többek</i> others	<i>között</i> among	<i>levest</i> soup-ACC	<i>tálalt fel.</i> served-3SG
49	17	exhaustive	‘For today’'s lunch, grandma served, among other things, soup.’ <i>Ami</i> which (relative)	<i>a ruhákat</i> the clothes-ACC	<i>illeti,</i> regards	<i>ez a szabó</i> this tailor	<i>csak</i> only	<i>ingeket</i> shirts-ACC	<i>foltozott be.</i> patched-3SG
50	17	unmodified	‘Regarding clothes, this tailor patched only shirts.’ <i>Ami</i> which (relative)	<i>a ruhákat</i> the clothes-ACC	<i>illeti,</i> regards	<i>ez a szabó</i> this tailor	<i>általában</i> usually	<i>ingeket</i> shirts-ACC	<i>foltozott be.</i> patched-3SG
51	17	contrastive	‘Regarding clothes, this tailor usually patched shirts.’ <i>Általában</i> usually	<i>a műhelyben</i> in the workshop	<i>ez a szabó</i> this tailor	<i>többek</i> others	<i>között</i> among	<i>ingeket</i> shirts-ACC	<i>foltozott be.</i> patched-3SG
52	18	exhaustive	‘Usually in the workshop this tailor patched, among other things, shirts.’ <i>Ami</i> which (relative)	<i>a táncot</i> the dance-ACC	<i>illeti,</i> regards	<i>Péter</i> Peter	<i>csak</i> only	<i>Marit</i> Mari-ACC	<i>kérte fel.</i> asked-3SG
53	18	unmodified	‘Regarding the dance, Peter asked only Mary.’ <i>Ami</i> which (relative)	<i>a táncot</i> the dance-ACC	<i>illeti,</i> regards	<i>Péter</i> Peter	<i>tegnap este</i> last night	<i>Marit</i> Mari-ACC	<i>kérte fel.</i> asked-3SG
54	18	contrastive	‘Regarding the dance, Peter asked Mary last night.’ <i>Tegnap este</i> last night	<i>a bálon</i> at the ball	<i>Péter</i> Peter	<i>többek</i> others	<i>között</i> among	<i>Marit</i> Mari-ACC	<i>kérte fel.</i> asked-3SG
			‘Last night at the ball, Peter asked, among others, Mary [for a dance].’						

References

- Balogh, Kata. 2007. Focus and 'Only' in Hungarian. In Balder D. Ten Cate and Henk W. Zeevat (eds.), *Logic, Language, and Computation* 4363, 31–44. Berlin, Heidelberg: Springer. https://doi.org/10.1007/978-3-540-75144-1_3 (accessed 2026 February 2).
- Balogh, Kata. 2013. Hungarian Pre-verbal Focus and Exhaustivity. In Yoichi Motomura, Alastair Butler, and Daisuke Bekki (eds.), *New Frontiers in Artificial Intelligence: JSAI-isAI 2012 Workshops, LENLS, JUR-ISIN, MiMI, Miyazaki, Japan, November 30 and December 1, 2012, Revised Selected Papers* (Lecture Notes in Computer Science 7856), 1–16. Berlin, Heidelberg: Springer. <https://doi.org/10.1007/978-3-642-39931-2> (accessed 2026 February 2).
- Comrie, Bernard and Haspelmath, Martin and Balthasar Bickel. 2015. The Leipzig Glossing Rules: Conventions for Interlinear Morpheme-by-Morpheme Glosses. Leipzig: Department of Linguistics of the Max Planck Institute for Evolutionary Anthropology; Department of Linguistics of the University of Leipzig.
- É. Kiss, Katalin (ed.). 1995. *Discourse Configurational Languages* (Oxford Studies in Comparative Syntax 4). New York: Oxford University Press.
- É. Kiss, Katalin. 1998. Identificational Focus versus Information Focus. *Language* 74(2), 245–273. <https://www.jstor.org/stable/417867> (accessed 2026 February 2).
- É. Kiss, Katalin. 1998. Multiple Topic, One Focus? *Acta Linguistica Hungarica* 45(1/2), 3–29. <https://www.jstor.org/stable/44656166> (accessed 2026 February 2).
- É. Kiss, Katalin. 1999. The English Cleft Construction as a Focus Phrase. In Lunella Mereu (ed.), *Boundaries of Morphology and Syntax* (Current Issues in Linguistic Theory 180), 217. Amsterdam; Philadelphia: John Benjamins Publishing Company. <https://doi.org/10.1075/cilt.180.14kis> (accessed 2026 February 2).
- É. Kiss, Katalin. 2007. Topic and Focus: Two Structural Positions Associated with Logical Functions in the Left Periphery of the Hungarian Sentence. *Acta Linguistica Hungarica* 55(3–4), 287–296. <https://doi.org/10.1556/aling.55.2008.3-4.4> (accessed 2026 February 2).
- Gécség, Zsuzsanna. 2020. Focused Topics in Hungarian: Reconsidering the Pragmatic Relevance of the Structural Focus Position. *Studia Linguistica* 74(3), 740–774. <https://doi.org/10.1111/stul.12142> (accessed 2026 February 2).
- Genzel, Susanne, Shinichiro Ishihara, and Balázs Surányi. 2015. The prosodic expression of focus, contrast and givenness: A production study of Hungarian. *Lingua* 165, 183–204. <https://doi.org/10.1016/j.lingua.2014.07.010> (accessed 2026 February 2).
- Gerőcs, Máttyás, Anna Babarczy, and Balázs Surányi. 2014. Exhaustivity in focus: Experimental evidence from Hungarian. In Joseph Emonds and Markéta Janebová (eds.), *Language Use and Linguistic Structure Proceedings of the Olomouc Linguistics Colloquium 2013*, 181–195. Olomouc, Czech Republic: Palacký University. <https://core.ac.uk/reader/42931210> (accessed 2026 February 2).
- Horvath, Julia. 2008. Separating “Focus Movement” from Focus. In Simin Karimi, Vida Samiian, and Wendy K. Wilkins (eds.), *Phrasal and Clausal Architecture: Syntactic Derivation and Interpretation. In Honor of Joseph E. Emonds*, 108–145. Amsterdam; Philadelphia: John Benjamins Publishing Company. <https://www.degruyterbrill.com/document/doi/10.1075/la.101.07hor/html> (accessed 2026 February 7).
- Hsu, Yu-Yin. 2019. Associations between Focus Constructions and Levels of Exhaustivity: An Experimental Investigation of Chinese. *PLOS ONE* 14(10), e0223502. <https://doi.org/10.1371/journal.pone.0223502> (accessed 2026 February 2).
- Hsu, Yu-Yin. 2024. Modal Raising and Focus Marking in Chinese. *Journal of Linguistics* 60(3), 635–666. <https://doi.org/10.1017/S0022226723000142> (accessed 2026 February 2).
- Káldi, Tamás and Anna Babarczy. 2016. A magyar fókusz és a skaláris implikaturák : egy szemmozgás-követéses kutatás eredményei [Hungarian focus and scalar implicatures: Results of an eye-tracking study]. In Bence Kas (ed.), *Szavard ne feledd!* 333–346. Budapest: MTA Nyelvtudományi Intézet. <https://real.mtak.hu/48340/> (accessed 2025 November 21).
- Káldi, Tamás. 2021. *Hungarian Pre-verbal Focus: Representation and Interpretation*. Ph.D. dissertation, Budapest University of Technology and Economics. <https://www.researchgate.net/publication/353145107> (accessed 2026 February 2).
- Káldi, Tamás; Szöllősi, Ágnes; Babarczy, Anna. 2021. Hungarian Structural Focus: Accessibility to Focused Elements and Their Alternatives in Working Memory and Delayed Recognition Memory. *Frontiers in Psychology* 12, 514886. <https://doi.org/10.3389/fpsyg.2021.514886> (accessed 2026 February 2).
- Kenesei, István. 2006. Focus as Identification. In Valéria Molnár and Susanne Winkler (eds.), *The Architecture of Focus*, 137–168. Berlin: Walter de Gruyter. (accessed 2026 February 2).

- Kornai, András; Kálmán, László. 1988. Hungarian sentence intonation. In Harry van der Hulst and Norval Smith (eds.), *Autosegmental studies on pitch accent*, 183–195. Dordrecht: Foris.
- Krifka, Manfred. 2008. Basic Notions of Information Structure. *Acta Linguistica Hungarica* 55(3–4), 243–276. <https://doi.org/10.1556/ALing.55.2008.3-4.2> (accessed 2026 February 2).
- Onea, Edgar. 2009. Exhaustiveness of Hungarian Focus: Experimental Evidence from Hungarian and German. In Arndt Riester and Edgar Onea (eds.), *Focus at the Syntax-Semantics Interface* (Working Papers of the SFB 732 3), 53–68. University of Stuttgart (accessed 2026 February 2).
- Onea, Edgar; Beaver, David. 2009. Hungarian Focus Is Not Exhausted. In *Proceedings of Semantics and Linguistic Theory (SALT 19)*, 342–359. <https://doi.org/10.3765/salt.v19i0.2524> (accessed 2026 February 2).
- Rooth, Mats. 1985. *Association with Focus*. Ph.D. dissertation, University of Massachusetts. <https://ecommons.cornell.edu/entities/publication/b16d4b65-30ea-4b87-9857-788aa69dc393> (accessed 2026 February 2).
- Rooth, Mats. 1992. A Theory of Focus Interpretation. *Natural Language Semantics* 1(1), 75–116. <https://doi.org/10.1007/BF02342617> (accessed 2026 February 2).
- Schwarz, Florian; Zehr, Jonathan. 2021. Tutorial: Introduction to PCIBex – An Open-Science Platform for Online Experiments: Design, Data-Collection and Code-Sharing. *Proceedings of the Annual Meeting of the Cognitive Science Society* 43, 15–16. <https://escholarship.org/uc/item/1ng1q4c6> (accessed 2026 February 2).
- Szabolcsi, Anna. 1981. Compositionality in Focus. *Folia Linguistica* 15(1–2), 141–161. <https://doi.org/10.1515/flin.1981.15.1-2.141> (accessed 2026 February 2).
- Vallduví, Enric; Vilks, Maria. 1998. On Rheme and Kontrast. In Peter Culicover and Louise McNally (eds.), *The Limits of Syntax*, 79–108. Leiden: Brill. https://doi.org/10.1163/9789004373167_005 (accessed 2026 February 2).
- Wedgwood, Daniel. 2006. Predication, Focus and the Positions of Negation in Hungarian. *Lingua* 116(3), 351–376. <https://doi.org/10.1016/j.lingua.2004.08.009> (accessed 2026 February 2).
- Wedgwood, Daniel; Petho, Gergely; Cann, Ronnie. 2006. Hungarian ‘Focus Position’ and English It-Clefts: The Semantic Underspecification of ‘Focus’ Readings. https://www.lel.ed.ac.uk/~ronnie/JoS-paper_forESRC.pdf (accessed 2026 February 2).
- Zimmermann, Malte. 2008. Contrastive Focus and Emphasis. *Acta Linguistica Hungarica* 55(3–4), 347–360. <https://doi.org/10.1556/ALing.55.2008.3-4.9> (accessed 2026 February 2).



Figure 12. Complete visual stimulus set with A) single- and B) multi-entity image pairs